

Reinforcement Learning-Based Decision Support for Just-in-Time Manufacturing Systems Under Stochastic Disruptions

1. Problem Statement

Just-in-Time (JIT) manufacturing systems are designed to minimize inventory and reduce waste; however, this efficiency increases vulnerability to supply chain disruptions. Traditional control strategies rely on static heuristics, such as inventory thresholds, to trigger actions like overtime. These approaches are inherently reactive and do not account for system dynamics or uncertainty.

This project investigates whether reinforcement learning (RL) can improve operational decision-making in a JIT environment by learning adaptive policies that reduce both shortages and unnecessary overtime.

2. System Description

The system modeled is a simplified dual-shift automotive production environment operating under JIT principles. The simulation includes stochastic demand and supply processes, where supply disruptions occur with a fixed probability.

Key system characteristics include:

- Two-shift production schedule
- Stochastic demand and supplier variability
- Shared inventory system
- Overtime as a controllable decision variable

The state space includes discretized inventory levels and a disruption indicator, allowing the RL agent to incorporate both system status and recent supply conditions.

3. Methodology

A Q-learning reinforcement learning algorithm was implemented to learn an optimal decision policy. At each time step, the agent selects between two actions:

- No action
- Call overtime

The reward function was designed to reflect operational tradeoffs:

- Shortages incur a high penalty
- Overtime incurs a smaller penalty
- Stable operation is rewarded

The RL agent was trained over **500 simulation episodes**. Performance was evaluated using **50 independent stochastic runs**, representing over 10,000 simulated production days.

A baseline reactive policy was implemented using a fixed inventory threshold for comparison.

All simulations were implemented in Python using NumPy and executed in a Jupyter/Colab environment.

4. Results

4.1 Performance Comparison

Metric	Reactive	RL
Reward	-16088.8 $\pm$ 2193.4	-13106.9 $\pm$ 2977.0
Shortage Rate	0.739 $\pm$ 0.096	0.611 $\pm$ 0.129
Overtime Rate	0.776 $\pm$ 0.093	0.627 $\pm$ 0.131

The RL policy improved total reward by approximately 3,000 points and reduced shortage frequency by approximately 17%. Additionally, overtime utilization was reduced by approximately 19%.

Notably, the RL policy improved both shortage frequency and overtime utilization simultaneously, indicating a dominant decision strategy rather than a tradeoff between competing objectives.

Figures 1, 2, and 3 below indicate that the RL agent successfully learned a more efficient and balanced decision strategy compared to the static threshold-based policy.

Figure 1. Total reward comparison between reactive and RL policies

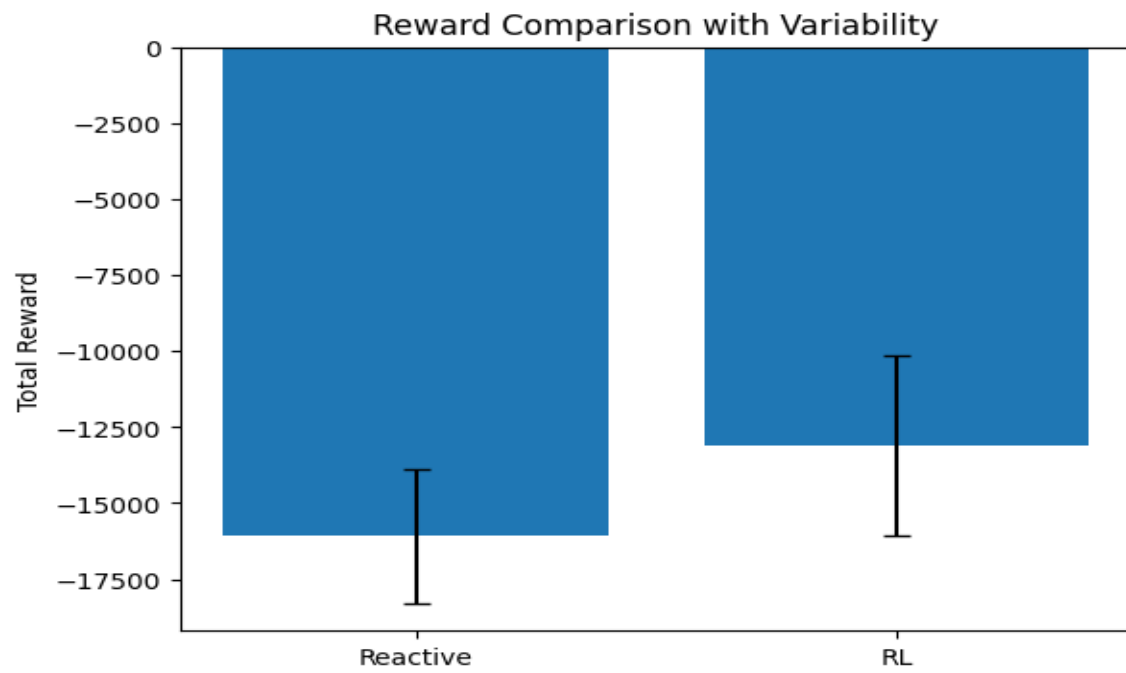


Figure 2. Shortage rate comparison between policies.

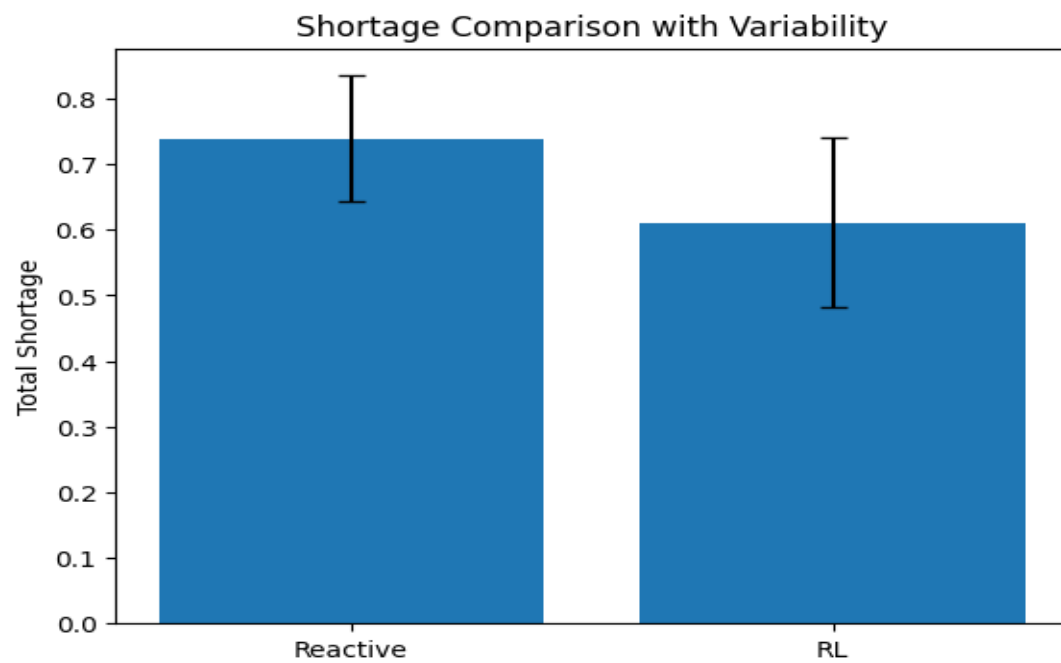
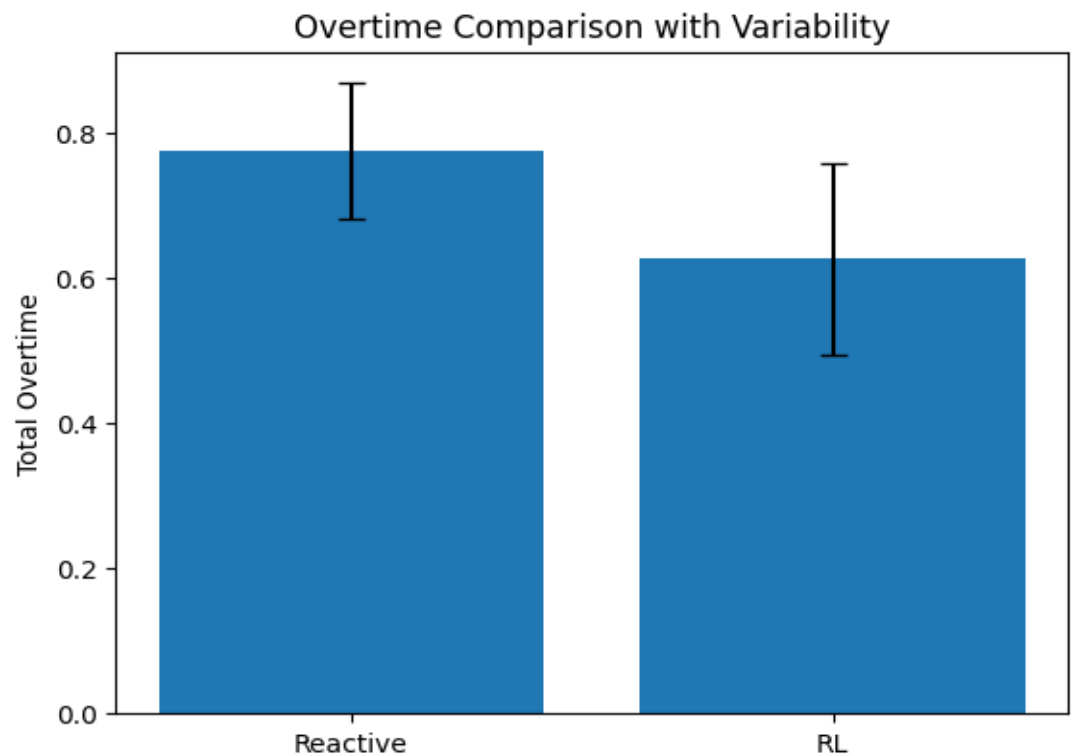


Figure 3. Overtime utilization comparison between policies.



4.2 Economic Impact

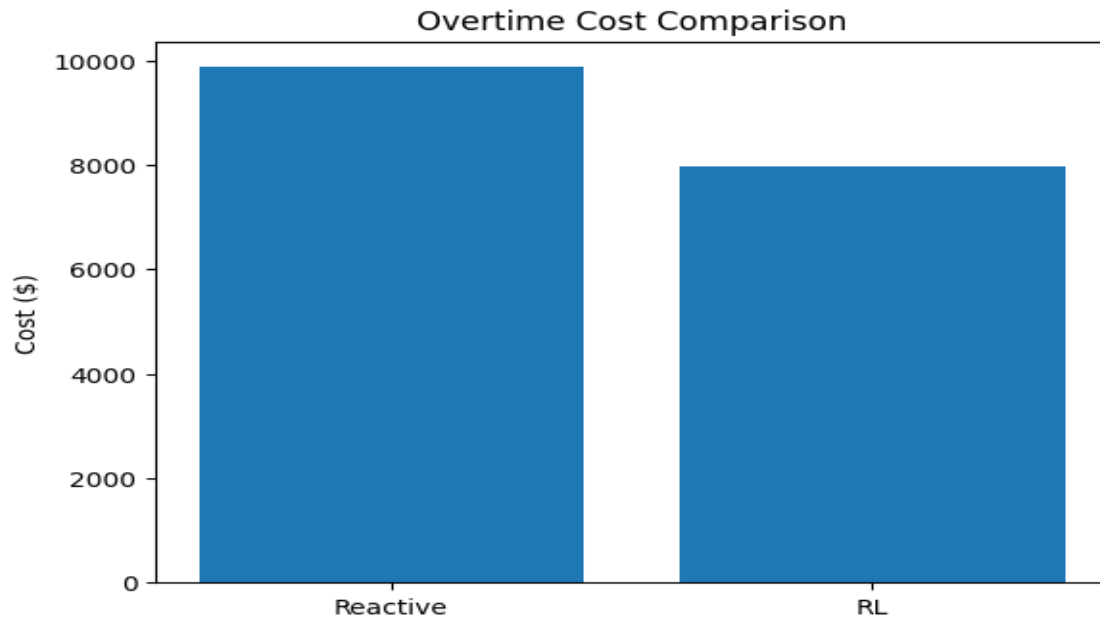
To quantify operational impact, overtime costs were estimated using an average labor rate of \$31.80 per hour.

Metric	Reactive	RL
OT Cost (200 days)	\$9,871.99	\$7,971.62
Savings	-	\$1,900.37

Annualized savings: **\$2,280.44 per year**

Figure 4 demonstrates that the RL policy not only improves system performance but also reduces operational cost.

Figure 4. Overtime cost comparison between policies.



5. Discussion

The RL agent demonstrated the ability to learn system dynamics and selectively apply interventions, avoiding unnecessary overtime while still mitigating shortage risk. Unlike the reactive policy, which triggers overtime based solely on inventory thresholds, the RL policy incorporates contextual information such as disruption conditions.

The results show that RL can simultaneously improve reliability (reduced shortages) and efficiency (reduced overtime), which is typically difficult to achieve with static control policies.

While variability in RL performance was higher across stochastic runs, the average improvement across all metrics indicates a robust advantage over the baseline approach.

6. Conclusion

This study demonstrates that reinforcement learning can be effectively applied to decision-making in JIT manufacturing systems. The RL-based policy improved system performance, reduced operational costs, and adapted to stochastic disruptions.

These findings suggest that data-driven, adaptive control strategies have strong potential to enhance resilience and efficiency in modern manufacturing environments.